

S.T.A.R. Research Charter

Scientific Specification for Arithmetic–Geometric–Topological Investigation of Cosmic Structure

Project: S.T.A.R. Program

Status: Research Charter v0.1 — Foundational Specification

Purpose: Canonical scientific specification for the S.T.A.R. research program

Primary domains: Arithmetic geometry, topology, topological data analysis, astrophysics, cosmology, statistical inference, mathematical physics, machine learning

1.1 Purpose

The S.T.A.R. Program investigates whether structures arising from arithmetic mathematics can be mapped into geometric and topological representations that exhibit statistically nontrivial, reproducible correspondence with independently measured astronomical and cosmological structure.

The program therefore operates across several layers:

```
[
\boxed{
\text{Arithmetic}
\rightarrow
\text{Geometry}
\rightarrow
\text{Topography}
\rightarrow
\text{Topology}
\rightarrow
\text{Observation}
\rightarrow
\text{Dynamics}
\rightarrow
\text{Physics}
}
]
```

The project does **not** begin by assuming that the universe is generated by elliptic curves.

Instead, elliptic curves and their arithmetic invariants constitute a candidate mathematical source space whose structural properties are to be investigated empirically.

The fundamental question is therefore:

[
\boxed{
\textbf{Can arithmetic invariants be mapped into geometric/topological structures in a way that
produces statistically nontrivial, reproducible correspondence with independent astronomical
data?}
}
]

This question is the canonical scientific identity of S.T.A.R.

All subsequent conjectures, mappings, computational experiments, and theoretical constructions shall be evaluated relative to this question.

1.2 What S.T.A.R. does not assume

The S.T.A.R. Program does not assume, as an established fact, that:

1. elliptic curves determine the structure of the universe;
2. elliptic-curve rank is identical to cosmic time;
3. rank is identical to gravitational potential;
4. arithmetic invariants replace the Einstein field equations;
5. dark matter is an arithmetic-curvature phenomenon;
6. dark energy is rank growth;
7. cosmic topology is necessarily identical to the topology of an arithmetic projection;
8. BSD is required for the central mapping;
9. a unique arithmetic-to-cosmic mapping exists;
10. a strict bijection exists between arithmetic objects and cosmic objects.

These remain hypotheses, candidate models, historical interpretations, or future theoretical possibilities unless independently established.

This distinction is essential because the existing ACSC material sometimes expresses these propositions as conclusions or axioms. The present Charter supersedes that presentation for purposes of scientific evaluation.

1.3 Scope

S.T.A.R. has two complementary objectives.

Objective A — Empirical investigation

Determine whether arithmetic structures produce measurable correspondence with astronomical structure.

Objective B — Theoretical development

If robust empirical correspondence exists, determine whether the correspondence can be expressed mathematically through:

- geometric mappings;
- topological constructions;
- variational principles;
- field theories;
- thermodynamic relations;
- dynamical equations;
- cosmological evolution models.

The theoretical program is therefore downstream of empirical validation wherever possible.

1.4 Historical continuity

The Charter does not invalidate the historical development of S.T.A.R.

Earlier concepts—including:

- ACSC;
- GLMPCT;
- ECC;
- RTCH;
- (Φ);
- rank-based elevation;
- regulator-based scaling;
- conductor/discriminant coordinates;
- Fibonacci and Lucas constructions;
- golden-ratio transformations;
- Heegner-point constructions;
- Virgo or other astronomical anchors;

- symbolic curvature;
- entropy fields;
- cohomological structures;

remain part of the project's historical research record.

Their status is changed where necessary from established conclusions to:

- candidate mappings;
- phenomenological models;
- conjectures;
- hypotheses;
- exploratory mathematical constructions;
- or future theoretical extensions.

Nothing is discarded merely because it has not yet been established.

2. EPISTEMIC LEVELS

A central requirement of S.T.A.R. is that mathematical derivation, empirical observation, modeling assumption, conjecture, and interpretation must never be presented as though they have identical evidentiary status.

The project therefore adopts the following epistemic hierarchy.

2.1 Level D — Mathematical Derivation

A statement is classified as **D** when it follows from explicitly stated mathematical definitions, assumptions, axioms, or previously established theorems.

Examples include:

[
 $F_Q = \Phi^* \omega$
]

when (Φ) and (ω) have been defined, and mathematical consequences derived from that definition.

A D-level result does not establish that the mathematical object corresponds to physical reality.

2.2 Level E — Empirical Observation

An **E** statement reports an observation obtained from data.

Examples:

- a measured galaxy-density distribution;
- an observed SFR distribution;
- a computed persistence diagram from an observational catalogue;
- a statistically measured correlation.

An empirical observation must identify:

- dataset;
 - sample;
 - population;
 - preprocessing;
 - measurement method;
 - uncertainty;
 - statistical procedure.
-

2.3 Level M — Model

An **M** statement defines a mathematical or computational model intended to represent a system.

Examples include:

$(\phi_E, \theta_E, z_E, s_E)$
]

or the historical candidate

[
 $z_E = \alpha r_E.$
]

The historical value

[
 $\alpha = 200$
]

is therefore retained as a model parameter rather than treated as a fundamental constant.

2.4 Level H — Hypothesis

An **H** statement proposes a relationship that is not yet established.

Examples:

```
[  
r_E \longleftrightarrow \text{cosmic-web complexity}  
]
```

or

```
[  
\operatorname{Reg}(E)  
\longleftrightarrow  
\text{void structure}.  
]
```

A hypothesis must have a corresponding experimental test.

2.5 Level C — Conjecture

A **C** statement proposes a broader mathematical or physical relationship whose truth remains unresolved.

ACSC, ECC, and related constructions may contain C-level propositions.

A conjecture becomes scientifically useful when it can be translated into explicit predictions.

2.6 Level T — Testable Prediction

A **T** statement specifies an observable result that could support or falsify a hypothesis.

For example:

A specified arithmetic mapping produces a persistence signature more similar to the observed cosmic web than its predefined null mappings, under a preregistered comparison metric.

A prediction must specify the statistic, comparison, population, and rejection criterion before the result is inspected whenever feasible.

2.7 Level S — Speculative Interpretation

An **S** statement proposes a possible interpretation without sufficient empirical or mathematical support.

Examples include:

- arithmetic ontology;
- symbolic universe interpretations;
- arithmetic origin of gravity;
- arithmetic explanations of dark matter or dark energy.

These may remain valuable research ideas, but they must be labeled as speculative.

3. THE FIVE CORE RESEARCH QUESTIONS

S.T.A.R. shall organize its research around five principal questions.

3.1 Core Question 1 — Arithmetic and Cosmic Structure

Question

Can arithmetic invariants be mapped into geometric/topological structures that correspond nontrivially to independently observed cosmic structure?

Formally, let

[
 $\mathcal{E}$
]

denote a selected family of arithmetic objects, initially elliptic curves, and let

[
 $I(E)$
]

denote an invariant vector such as

$\left(\begin{array}{l} r_E, \\ \log |\Delta_E|, \\ \log N_E, \\ \log \operatorname{Reg}(E), \\ T_E, \\ \vdots \end{array} \right).$

Define a mapping

[
 $\Phi: \mathcal{E} \rightarrow \mathbb{R}^d.$
]

The question is whether

[
 $\Phi(\mathcal{E})$
]

possesses geometric or topological properties that correspond to independently obtained astronomical data.

The correspondence must be evaluated against null and alternative constructions.

3.2 Core Question 2 — Arithmetic Topography

Question

Can arithmetic invariants be converted into scalar or vector fields whose geometry can be compared with astronomical fields?

A general arithmetic topography may be represented by

$$f!\left(\begin{array}{l} I_1(E), \\ I_2(E), \\ \vdots, \\ I_n(E) \end{array} \right),$$

where (f) is an explicitly defined mapping.

From this one may define, where mathematically appropriate,

$$-\nabla h_{\rm arith}.$$

The existence of such a mathematical field does not imply that it is gravity.

The physical question is instead whether the field exhibits measurable correspondence with:

- matter density;
- gravitational potential;
- weak lensing;
- peculiar velocity;
- radiation;
- galaxy morphology;
- star formation;
- other independently measured observables.

3.3 Core Question 3 — Rank and Cosmic-Web Morphology

Rank is one candidate arithmetic organizing variable.

The principal hypothesis family is:

$$[r_E \longleftrightarrow$$

$C_{\rm cosmic}$,
]

where ($C_{\rm cosmic}$) represents measurable cosmic-web structure.

The initial investigation shall consider:

- clusters;
- filaments;
- voids;
- connectivity;
- local density;
- topological complexity.

Rank must be compared against other invariants rather than assumed to be uniquely responsible.

At minimum:

[
 r_E ,
 $\operatorname{Reg}(E)$,
 N_E ,
 $|\Delta_E|$,
 T_E
]

should be evaluated individually and jointly.

A composite invariant vector may be written:

[
 $X_E =$
 $\left($
 $r_E,$
 $\log \operatorname{Reg}(E),$
 $\log N_E,$
 $\log |\Delta_E|,$
 T_E
 $\right).$
]

3.4 Core Question 4 — Cosmic Evolution and Persistent Homology

Question

Can the evolution of cosmic topology with redshift be related to an independently defined evolution of arithmetic structure?

Observed cosmic structure may be represented as

$$[D_{\text{cosmic}}(z),]$$

where (z) is redshift.

An arithmetic family may instead be indexed by an abstract evolution parameter

$$[\tau,]$$

giving

$$[D_{\text{arith}}(\tau).]$$

The research question is whether there exists an empirically supported correspondence

$$[\tau=f(z)]$$

such that

$$[D_{\text{arith}}(\tau)]$$

reproduces relevant features of

$$[D_{\text{cosmic}}(z).]$$

The project shall **not assume**

[
 $\tau=z$
]

or

[
 $\tau=r.$
]

Redshift is observational ordering; it is not automatically equivalent to arithmetic time.

Persistent homology is particularly useful because it can distinguish changes in topology from simple metric rescaling.

3.5 Core Question 5 — Symbolic Fields and Thermodynamic/Energy Relations

The deepest theoretical question is whether the arithmetic/topological framework can eventually generate physically meaningful fields and thermodynamic relations.

This includes investigation of:

- symbolic fields;
- entropy;
- cohomology;
- geodesics;
- scalar coupling;
- phase transitions;
- mass-energy relations;
- variational principles;
- quantum-scale extensions.

A generalized energy relation may ultimately take a form schematically represented by

E_{rest}
+
 E_{field}
+
 $E_{\text{topological}}$
+

$E_{\rm entropy}$
 $+\cdots,$
 $]$

with

$[$
 $E_{\rm rest}=mc^2$
 $]$

recovered as an appropriate limiting case if the theory is viable.

No such generalized relation shall be treated as established until dimensional consistency, mathematical derivation, conservation laws, known physical limits, and independent empirical predictions have been demonstrated.

4. DEFINITIONS OF KEY OBJECTS

4.1 Arithmetic object

The initial arithmetic object is an elliptic curve over $(\mathbb{Q})$:

$[$
 $E/\mathbb{Q}.$
 $]$

The project may later investigate broader classes of arithmetic objects.

4.2 Arithmetic invariants

Candidate invariants include:

Rank

$[$
 $r_E.$
 $]$

Conductor

[
N_E.
]

Discriminant

[
\Delta_E.
]

Regulator

[
\operatorname{Reg}(E).
]

Torsion

[
T_E=E(\mathbb Q)_{\rm tors}.
]

Additional quantities may include:

- Frobenius traces;
- (j)-invariant;
- periods;
- Heegner-point heights;
- Selmer ranks;
- L-function quantities.

No invariant is assumed to have physical meaning before testing.

4.3 The projection (\Phi)

A central candidate mapping is

(\phi_E,\theta_E,z_E,s_E).
]

Historically, candidate coordinates include normalized functions of

$$[\log|\Delta_E|, \\ \log N_E, \\ r_E, \\ \operatorname{Reg}(E).]$$

A simplified historical form is

$$(\log|\Delta_E|, \\ \log N_E, \\ \alpha r_E).]$$

The project explicitly adopts the interpretation that (Φ) is a **projection**, not necessarily a bijection.

Multiple arithmetic objects may therefore map into the same or nearby geometric region.

This is compatible with the concept of symbolic fibers or equivalence classes.

4.4 Candidate mapping families

The project shall maintain multiple mapping families.

Family (Φ)

The principal historical arithmetic projection.

MCJ

A (j) -invariant-based mapping.

PTD

A purely algebraic mapping independent of analytic L-function assumptions.

FT

A Frobenius-trace-based mapping.

The existing alternative-mapping document explicitly emphasizes that the program should remain viable independently of BSD.

These alternatives are not merely optional demonstrations.

They are essential controls against the possibility that one mapping produces apparent correspondence simply because its degrees of freedom were selected after observing the astronomical data.

4.5 Arithmetic topography

Let

[
 $h_{\{\text{rm arith}\}:\Omega\rightarrow\mathbb{R}}$
]

be a scalar field generated from arithmetic data.

Possible constructions include:

$f(r, \operatorname{Reg}, N, \Delta, T),$
]

or spatial interpolation of discrete arithmetic samples.

The resulting topography can be studied through:

- gradients;
- critical points;
- curvature;
- level sets;
- ridges;
- basins;
- density;
- persistence.

Physical interpretation must remain separate from mathematical construction.

4.6 Arithmetic topology

Given a point cloud

[
 $P_{\Phi} = \Phi(\mathcal{E}) \subset \mathbb{R}^d,$
]

construct a filtration

[
 $K_{\{\epsilon_1\}}$
 $\subseteq$
 $K_{\{\epsilon_2\}}$
 $\subseteq$
 $\cdots$
]

Persistent homology then produces

[
 $D_k(P_{\Phi})$
]

for homological dimensions ($k=0,1,2,\dots$).

The existing ACSC documents propose this general pipeline and comparison with observational point clouds.

4.7 Interpretation of Betti numbers

The following interpretation shall be used cautiously.

(β_0)

Connected components.

This can be compared with cluster-like connected structure.

(β_1)

One-dimensional loops/connectivity structures.

(β_1) shall **not** automatically be interpreted as "number of filaments."

Filament identification requires geometric or graph-based analysis in addition to persistent homology.

(β_2)

Two-dimensional cavities.

These may be compared with void-like structures.

Thus:

```
[  
  \beta_0  
  \rightarrow  
  \text{component structure},  
]
```

```
[  
  \beta_1  
  \rightarrow  
  \text{connectivity/loop structure},  
]
```

```
[  
  \beta_2  
  \rightarrow  
  \text{cavity structure}.  
]
```

The mapping from these mathematical quantities to named astronomical structures remains empirical.

4.8 Topological comparison

Persistence diagrams may be compared using distances such as

[
W_p(D₁,D₂)
]

or bottleneck distance

[
d_B(D₁,D₂).
]

The existing ACSC work uses Wasserstein distance as a principal comparison measure.

However, a numerical threshold such as

[
W₂<10⁻²
]

shall not automatically constitute evidence for the theory.

Thresholds must be calibrated against:

- sample size;
- observational noise;
- normalization;
- resolution;
- null models;
- alternative mappings;
- dimensionality;
- parameter fitting.

5. RANK–COSMIC-WEB HYPOTHESES

The rank program shall be divided into explicit hypotheses rather than one broad assertion.

H₁ — Rank–Cluster Hypothesis

There exists a statistically significant relationship between arithmetic rank and cluster-like cosmic structure.

Possible observables include:

[
 ρ_{galaxy} ,
]

cluster membership,

local density,

or persistent (β_0) structure.

The hypothesis does not specify the direction of the relationship in advance unless a directional prediction is formally preregistered.

H₂ — Rank–Connectivity Hypothesis

Arithmetic rank is statistically associated with large-scale connectivity structures.

Possible observables include:

- graph connectivity;
- skeleton structure;
- persistence of (β_1) features;
- filament length;
- node degree;
- network centrality.

Persistent (β_1) is therefore one measurement rather than the complete definition of a filament.

H₃ — Rank–Void Hypothesis

Arithmetic rank or combinations of arithmetic invariants are associated with cavity or underdense cosmic regions.

Possible observables include:

- void volume;
- void radius;
- void density;

- (β_2) persistence;
- distance from high-density structures.

H₄ — Rank–Web Complexity Hypothesis

Arithmetic rank contributes measurable information about total cosmic-web complexity.

A candidate complexity measure may combine:

```
f(
  \beta_0,
  \beta_1,
  \beta_2,
  \text{persistence},
  \text{graph structure},
  \text{density}
).
```

The question is whether rank explains independent variance after controlling for known astronomical variables.

H₅ — Invariant Competition Hypothesis

Rank is not evaluated in isolation.

The project shall test whether:

```
[
  r_E
]
```

outperforms, matches, or is subsumed by other invariants as predictors.

Candidate comparisons include:

```
[
  r_E
  \quad \text{vs.} \quad
  \operatorname{Reg}(E),
]
```

[
r_E
\quad\text{vs.}\quad
N_E,
]

[
r_E
\quad\text{vs.}\quad
|\Delta_E|,
]

and

[
r_E
\quad\text{vs.}\quad
X_E.
]

The scientific objective is not to establish rank as the winner but to determine which arithmetic structures, if any, carry reproducible information.

6. STATISTICAL AND NULL-MODEL REQUIREMENTS

This section is mandatory for every empirical claim.

6.1 Independent data

Arithmetic data and astronomical data should be independently generated wherever practical.

The mapping must not be tuned against the same observational sample used for final validation.

6.2 Null models

Every positive result must be compared against appropriate null models.

Examples include:

1. random permutations;
2. randomized invariant assignments;
3. spatially randomized point clouds;
4. shuffled rank values;
5. alternative arithmetic sequences;
6. alternative coordinate mappings;
7. isotropic controls;
8. matched-density controls;
9. bootstrap controls;
10. simulation-derived controls.

6.3 Alternative mappings

At least one family of alternative mappings must be evaluated.

The existing MCJ, PTJ, and FT proposals provide an initial framework.

This is necessary to distinguish:

[
 \text{arithmetic information}
]

from

[
 \text{generic geometric flexibility}.
]

6.4 Parameter locking

Parameters used in the mapping must be classified as either:

- fixed before analysis;
- derived mathematically;
- fitted on training data;
- or exploratory.

A parameter selected because it improves the final observational correspondence cannot subsequently be described as a prediction.

6.5 Train/validation/test separation

When machine learning or symbolic regression is used:

```
[  
  \text{training}  
  \rightarrow  
  \text{validation}  
  \rightarrow  
  \text{blind test}  
]
```

shall be maintained.

The test set must not be used to choose:

- mapping parameters;
 - features;
 - model architecture;
 - symbolic equations;
 - thresholds.
-

6.6 Multiple hypothesis correction

If many candidate invariants, mappings, redshift bins, topological dimensions, or statistics are tested, the analysis must account for multiplicity.

Potential methods include:

- false-discovery-rate control;
 - family-wise error control;
 - permutation-based global testing;
 - preregistered primary endpoints.
-

6.7 Effect size

Statistical significance alone is insufficient.

Every result should report:

- effect size;
 - uncertainty;
 - confidence or credible interval where appropriate;
 - sample size;
 - null distribution;
 - practical magnitude;
 - robustness.
-

6.8 Out-of-sample replication

A candidate relationship is not considered established because it works on one dataset.

A strong result should survive:

```
[  
\text{dataset A}  
\rightarrow  
\text{dataset B}  
\rightarrow  
\text{independent replication}.  
]
```

6.9 Simulation controls

Where possible, observational results should be compared with cosmological simulations.

This helps determine whether an apparent arithmetic correspondence is actually responding to:

- ordinary clustering;
 - selection effects;
 - survey geometry;
 - sampling density;
 - redshift distortions;
 - known cosmic-web statistics.
-

7. CLAIM CLASSIFICATION AND REGISTRY LOGIC

Every substantive S.T.A.R. claim shall receive a registry entry.

A minimum claim record shall contain:

Field	Requirement
Claim ID	Unique identifier
Statement	Exact claim
Category	D/E/M/H/C/T/S
Source	Document or experiment
Mathematical basis	Required where applicable
Dataset	Required for empirical claims
Prediction	Required for hypotheses
Null model	Required for empirical hypotheses
Status	Open / supported / contradicted / unresolved
Reproducibility	Code/data status
Dependencies	Other claims required
Last tested	Version/date

7.1 Example classifications

Historical rank-elevation mapping

```
[  
z=200r  
]
```

Classification:

[
 M
]

unless a future experiment establishes independent predictive support.

(Φ) projection

($\log|\Delta|, \log N, \alpha r$)
]

Classification:

[
 M
]

Rank corresponds to cosmic complexity

Classification:

[
 H
]

with associated experiments.

Arithmetic projection reproduces cosmic topology

Classification:

[
 $\mathrm{C/H}$
]

until independently demonstrated.

Arithmetic field represents gravity

Classification:

[
 H
]

Arithmetic replaces the Einstein field equation

Classification:

[
 S
]

unless a complete mathematically and empirically validated theory is eventually developed.

ECC entropy/cohomology relation

Classification:

[
 $\mathrm{C/M}$
]

until formal mathematical proof and independent empirical validation justify stronger classification.

8. RELATIONSHIP OF EXISTING S.T.A.R. DOCUMENTS

The Research Charter becomes the top-level specification.

Existing documents become subordinate research modules.

8.1 ACSC

The Arithmetic–Cosmic Structure Conjecture becomes the principal structural hypothesis.

Its core contribution is the proposed relationship

[

$$\overset{E}{\Phi} \longrightarrow M_{\text{arithmetic}}$$

]

and the comparison of the resulting topology with cosmic structure.

The existing ACSC monograph contains stronger claims—including a proposed topological embedding lemma and specific Wasserstein thresholds—that shall now be treated as candidate propositions or experimental criteria rather than established theorems unless independently proved.

ACSC therefore becomes:

[

$$\boxed{\text{Primary structural hypothesis}}$$

]

rather than an assumed description of reality.

8.2 GLMPCT

GLMPCT becomes the mapping and scale-translation framework.

Its principal role is to investigate how local arithmetic information can be projected into global geometric representations.

The Φ mapping belongs primarily here.

GLMPCT shall therefore address:

- coordinate construction;
- normalization;
- scale;

- projection;
- equivalence classes;
- degeneracy;
- alternative mappings.

The existing alternative-mapping document explicitly states that strict bijection is not required.

This principle is retained.

8.3 ECC

The Entropy Cohomology Conjecture becomes a candidate dynamical/thermodynamic layer.

Its purpose is to investigate whether topological and cohomological structure can constrain complexity, entropy, and dynamical evolution.

The existing arithmetic-cartography document describes ECC as a dynamical constraint on ACSC and introduces a symbolic entropy field and cohomological structure.

Under this Charter, those constructions remain candidate mathematical structures.

They are not yet accepted as physical laws.

8.4 RTCH

RTCH — the proposed Relational/Topological Cohomological framework — becomes a candidate variational completion.

Its role is downstream of the empirical program.

A viable RTCH model must eventually demonstrate:

1. a well-defined action;
2. mathematically consistent field equations;
3. conservation laws;
4. correct known limits;
5. stability;
6. well-posed evolution;
7. observational constraints;
8. novel testable predictions.

RTCH must therefore not be used to justify ACSC before ACSC has independent empirical support.

8.5 Historical phenomenological constructions

The following remain in the project archive:

- $(z=200r)$;
- Fibonacci/Lucas mappings;
- golden-ratio scaling;
- regulator scaling;
- Heegner constructions;
- (π) -based angular constructions;
- astronomical anchor points;
- symbolic curvature;
- rank elevation.

They shall be organized under:

```
[  
\boxed{\text{Candidate Mapping Registry}}  
]
```

and tested against alternatives.

9. PHASED RESEARCH ROADMAP

Phase 0 — Scientific Foundation

Establish:

1. Research Charter;
2. Claim/Evidence Registry;
3. terminology registry;
4. mapping registry;
5. parameter registry;
6. data provenance requirements.

Deliverable: frozen Charter v1.0.

Phase 1 — Document Reconciliation

Every existing document shall be mapped against the Charter.

For every major statement:

```
[
\text{Existing claim}
\rightarrow
\text{Charter category}
\rightarrow
\text{required evidence}
\rightarrow
\text{current status}.
]
```

No historical research is deleted.

Instead:

```
[
\text{claim}
\rightarrow
\text{reclassified claim}.
]
```

Phase 2 — Arithmetic Dataset

Construct a reproducible arithmetic database containing, where available:

```
[
E,\quad
N_E,\quad
\Delta_E,\quad
r_E,\quad
\operatorname{Reg}(E),\quad
T_E,
]
```

plus relevant additional invariants.

Record:

- source;
 - computational method;
 - software;
 - version;
 - precision;
 - verification status.
-

Phase 3 — Mapping Benchmark

Construct a common benchmark for:

```
[  
  \Phi, \quad  
  \mathrm{MCJ}, \quad  
  \mathrm{PTD}, \quad  
  \mathrm{FT},  
]
```

plus null mappings.

Every mapping receives:

- equations;
 - parameter list;
 - normalization;
 - implementation;
 - version;
 - expected output.
-

Phase 4 — Arithmetic Topography

Construct candidate fields:

```
[  
  h_{\rm arith}(\mathbf{x}).  
]
```

Investigate:

- gradients;
- density;
- critical points;
- ridges;
- basins;
- curvature;
- topological signatures.

At this stage the objective is mathematical characterization, not physical interpretation.

Phase 5 — Cosmic-Web Experiment

Construct matched astronomical datasets.

Candidate sources include galaxy surveys and simulations.

Compute:

```
[
D_0,\quad D_1,\quad D_2.
]
```

Compare arithmetic and astronomical structures using:

- Wasserstein distance;
 - bottleneck distance;
 - persistence landscapes;
 - persistence images;
 - graph-based morphology;
 - spatial statistics.
-

Phase 6 — Rank Experiment

Test:

```
[
H_1,H_2,H_3,H_4,H_5.
]
```

Compare rank against other arithmetic invariants.

This becomes the first major falsifiable empirical program.

Phase 7 — Cosmic Evolution

Repeat the topology analysis across redshift.

Construct:

[
 $D_{\text{cosmic}}(z)$
]

and

[
 $D_{\text{arith}}(\tau)$
]

Test whether an independently supported mapping

[
 $\tau=f(z)$
]

exists.

Phase 8 — Astrophysical Laboratory

Investigate whether arithmetic/topological variables predict:

- star formation rate;
- stellar mass;
- gas mass;
- metallicity;
- morphology;
- environment.

Machine learning may be used for feature discovery.

However:

```
[
\boxed{
\text{ML discovery}\neq\text{proof}
}
]
```

The correct pipeline is:

```
[
\text{ML discovery}
\rightarrow
\text{candidate relation}
\rightarrow
\text{independent validation}
\rightarrow
\text{analytic investigation}.
]
```

Phase 9 — Physical Correspondence

Only after robust structural correspondence has been demonstrated should the program investigate:

- gravitational fields;
 - weak lensing;
 - radiation;
 - peculiar velocities;
 - scalar fields;
 - cosmological expansion.
-

Phase 10 — Theoretical Dynamics

Investigate:

- ACSC;
- GLMPCT;
- ECC;
- RTCH;
- action principles;
- field equations;

- conservation;
 - thermodynamics;
 - energy relations.
-

Phase 11 — Cosmological Application

Only after a consistent dynamical theory exists should the program investigate:

- expansion history;
- Hubble measurements;
- early-universe constraints;
- Planck-derived constraints;
- Hubble–Planck tension.

The Hubble–Planck tension is therefore a **downstream validation problem**, not the starting target of the theory.

10. FALSIFICATION CRITERIA

S.T.A.R. must define failure conditions before interpreting positive results.

10.1 Structural failure

ACSC is substantially weakened if independently selected arithmetic mappings consistently fail to reproduce any nontrivial correspondence with observed cosmic topology.

10.2 Null indistinguishability

If arithmetic mappings perform no better than randomized or generic geometric controls, the arithmetic hypothesis loses empirical support.

10.3 Mapping dependence

If apparent correspondence exists only for one highly tuned mapping while reasonable alternative mappings and null models fail, the result shall be classified as mapping-dependent rather than general evidence for arithmetic structure.

10.4 Sample instability

If correspondence disappears when:

- sample size changes;
- survey region changes;
- redshift range changes;
- resolution changes;
- independent datasets are used;

then the result shall not be considered robust.

10.5 Invariant instability

If an observed relationship depends entirely upon one arbitrary arithmetic invariant without surviving alternative-invariant testing, the interpretation must remain limited to that invariant and cannot automatically be generalized to arithmetic structure as a whole.

10.6 Predictive failure

A candidate theory must be considered unsuccessful if it cannot make predictions that distinguish it from existing models.

10.7 Physical inconsistency

Any proposed physical theory must be rejected or revised if it violates essential requirements such as:

- dimensional consistency;

- conservation laws;
- causality;
- stability;
- known experimental limits;
- well-posedness;
- established low-energy or classical limits.

10.8 The strongest falsification

The strongest test is prospective prediction.

A mature S.T.A.R. result should take the form:

```
[
\boxed{
\text{Arithmetic structure}
\rightarrow
\text{fixed mapping}
\rightarrow
\text{quantitative prediction}
\rightarrow
\text{independent observation}
}
]
```

where the final observation was not used to construct the mapping.

11. SCIENTIFIC STANDARD OF SUCCESS

The project shall distinguish several levels of success.

Level 0 — Mathematical construction

A coherent mapping can be defined.

Level 1 — Computational reproducibility

Independent researchers can reproduce the mapping and calculations.

Level 2 — Statistical correspondence

The mapping exhibits statistically nontrivial correspondence with astronomical data.

Level 3 — Cross-dataset replication

The correspondence survives independent datasets.

Level 4 — Predictive correspondence

The mapping predicts previously unused observations.

Level 5 — Physical theory

A mathematically consistent dynamical theory derives the correspondence and produces new physical predictions.

S.T.A.R. shall not describe a Level 1 or Level 2 result using the language appropriate to Level 5.

12. GOVERNING PRINCIPLE

The central methodological principle of the S.T.A.R. Program is:

```
[
\boxed{
\text{Do not ask whether the theory can be made to fit the universe.}
}
]
```

Instead:

```
[
\boxed{
\text{Ask whether a fixed mathematical construction predicts structure that was not used to construct it.}
}
]
```

The ultimate objective is therefore not confirmation of ACSC, ECC, GLMPCT, rank, or any particular mapping.

The objective is to determine whether the data support a reproducible relationship between arithmetic structure and cosmic structure.

If the relationship exists, the program should progressively formalize it.

If it does not exist, the program should identify where and why the proposed correspondence fails.

Either outcome is scientifically informative.

13. CANONICAL RESEARCH PIPELINE

The entire S.T.A.R. program may be represented as:

```
[  
  \boxed{  
    E  
    \rightarrow  
    I(E)  
    \rightarrow  
    \Phi(E)  
    \rightarrow  
    \mathcal{H}_{\rm arith}  
    \rightarrow  
    D_{\rm arith}  
    \rightarrow  
    D_{\rm cosmic}  
    \rightarrow  
    \text{Statistical Test}  
    \rightarrow  
    \text{Independent Validation}  
  }  
]
```

where:

- E is the arithmetic object;
- $I(E)$ is its invariant vector;
- $\Phi(E)$ is the candidate geometric projection;
- $\mathcal{H}_{\rm arith}$ is the resulting arithmetic topography/topology;
- $D_{\rm arith}$ is its topological signature;
- $D_{\rm cosmic}$ is the observational topological signature.

Only after this pipeline produces robust evidence should the program proceed to:

```
[  
  \boxed{  
    \text{Correspondence}  
    \rightarrow  
    \text{Dynamics}  
    \rightarrow  
    \text{Physical Theory}.  
  }  
]
```

This establishes the Research Charter as the governing scientific framework for all subsequent S.T.A.R. development.